School of Computer Sciences, Universiti Sains Malaysia

CCS516 Computational Intelligence

Semester 1, 2026/2027

Assignment 01

Solving Traveling Salesman Problem (TSP) Using Distributed Evolutionary Algorithms in Python (DEAP)

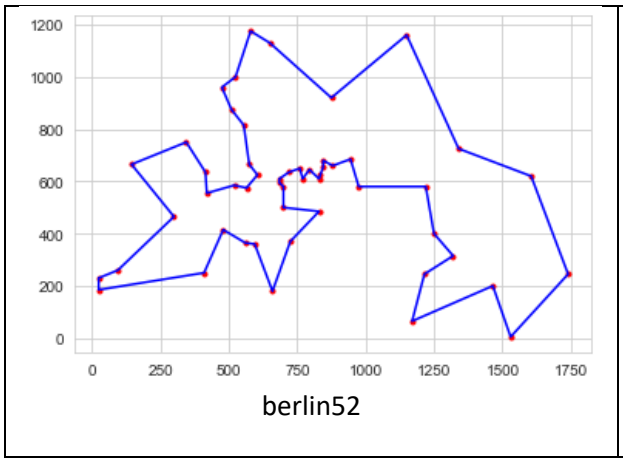
The Traveling Salesman Problem (TSP) is an NP-hard problem in combinatorial optimization where a salesman must determine the shortest route that visits each point in a given set exactly once.

Design an Evolutionary Algorithm (EA) to solve the following TSPs, i.e., berlin52, eil101, tsp225, pcb442, and pr1002, which the size of each TSP is 52, 101, 225, 442, and 1002 cities respectively.

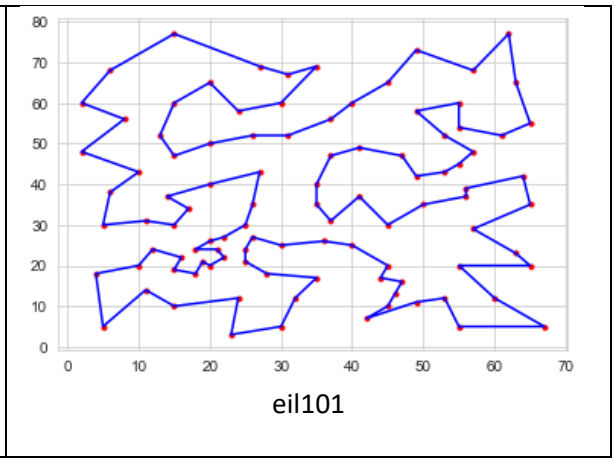
Three source files are given as follows: `solve_tsp.py`, `tsp.py`, and `elitism.py`. You are required to design an EA based on the three source files given. In other words, the given three source files serve as a foundation for your EA design.

1. Use `tsp.py` to generate a plot for the optimal tour specified in `berlin52.opt.tour`, `eil101.opt.tour`, `tsp225.opt.tour`, `pcb442.opt.tour`, and `pr1002.opt.tour`. Refer to Figure 1 for the plots. The city coordinates for the five TSPs are given as follows, `berlin52.tsp`, `eil101.tsp`, `tsp225.tsp`, `pcb442.tsp`, and `pr1002.tsp`.

2. Modify `tsp.py` such that it finds the optimal tour length (rounded to the nearest integer) for berlin52, eil101, tsp225, pcb442, and pr1002.



berlin52



eil101

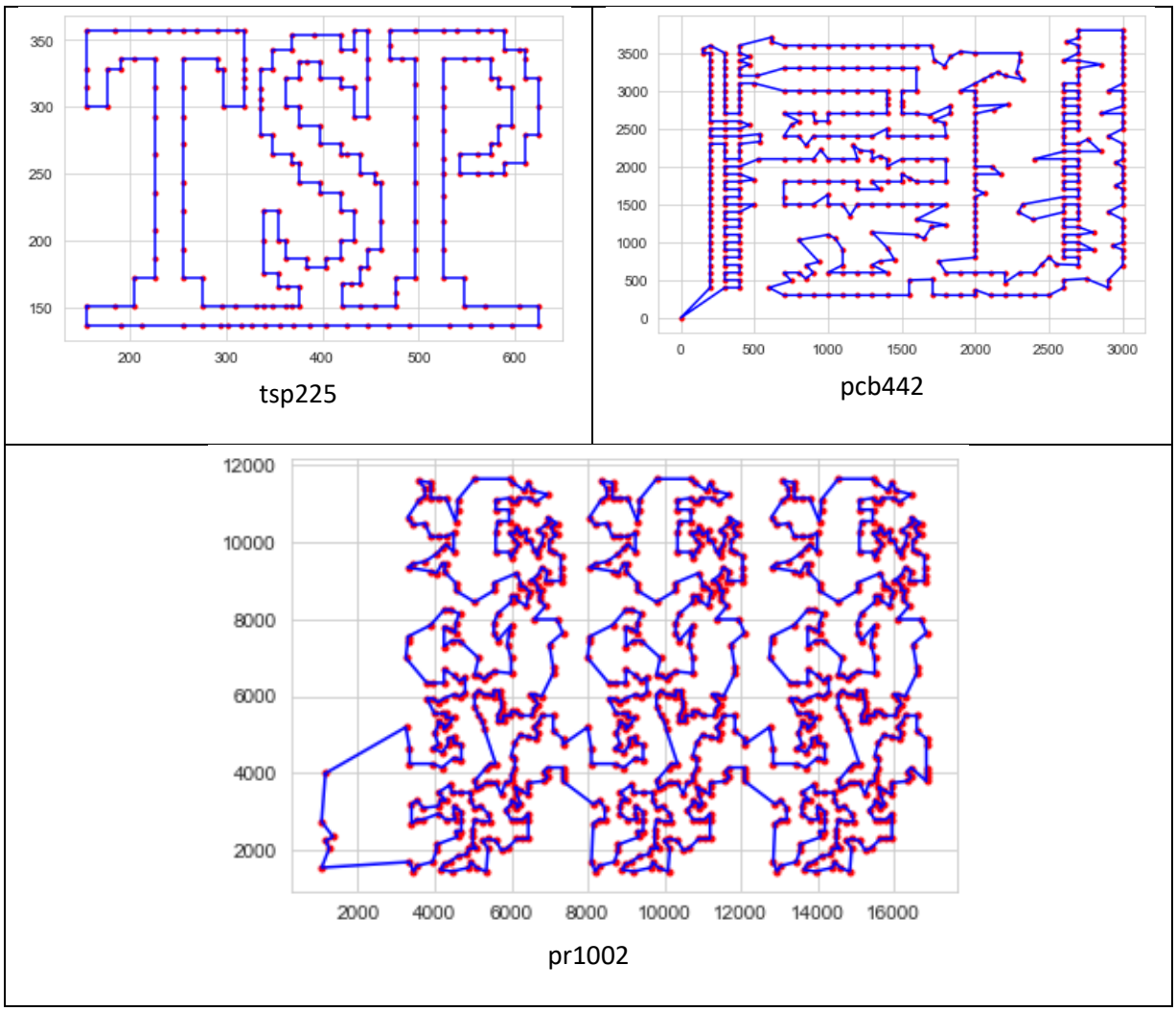


Figure 1: Plot of optimal tour for berlin52, eil101, tsp225, pcb442, and pr1002.

3. Modify `solve_tsp.py` and `elitism.py` to design an evolutionary algorithm (EA) to solve berlin52, eil101, tsp225, pcb442, and pr1002. Except for the representation scheme and objective function, your EA design should cover the following component: variation operators and their respective probability, selection (parent and survivor), parents and offspring population management models (e.g., population size), initialization, random seed for each run, and termination condition. Limit each run to a maximum of 50,000 fitness function evaluations, and report the results averaged over 5 runs. Also, the deviation percentage of the average tour length from the optimal tour length must be computed and reported.

Assignment due date: 22 Nov 2026 (Sunday), 11:59 p.m. (Week 08).

Assignment type: Pair assignment (a group of two members). The member grouping will be done via the eLearn@USM.

Submission procedure: A zip/rar package which consists of the following items is to be submitted via the eLearn@USM:

1. An assignment report (Times New Roman, font size 12, single spacing, not more than **15** pages) which explains the algorithm and technique used as well as the experiments performed to obtain the best results. Except for the representation scheme and objective function, all the important aspects such as variation operators and their respective probability, selection (parent and survivor),

parents and offspring population management models (e.g., population size), initialization, random seed for each run, and termination condition, must be included. [Hint to create a good report: The report should be prepared in such a way that it allows one to replicate your experiments]

2. A cover page which contains your details is expected to be included in your assignment report.
3. All other supporting documents which can make the evaluator understand your work (e.g. source codes, screenshots, additional files from the relevant tools used in implementing the assignment).

The zip/rar package must be named according to the following notation: **CCS516_[GroupNumber]_A01**. For example, for Group03, they must name the zip/rar package as CCS516_Group03_ A01.

Reference: Kindly state any source of reference should you refer to various sources to complete this assignment.

Grading rubric:
The total mark for this project is 100 marks. The total will be scaled to 8% of your overall grade. It will consider the following:

1. Plots for optimal tour and the respective tour length (30%): Generate the plots for optimal tour and compute the respective tour length.
2. Algorithm design (40%): Describe and implement the technique/algorithm that are used to solve the TSP.
3. Experimental results (20%): Explain the results and show how the results are used to support decisions.
4. Problem and pitfalls (10%): Discuss the mistakes that have been made, and the knowledge & experience gained throughout the assignment implementation.

	Very Weak (1 – 2 points)	Weak (3 – 4 points)	Fair (5 – 6 points)	Good (7 – 8 points)	Very Good (9 – 10 points)
Plots for optimal tour and the respective tour length (30%)	• Able to generate optimal tour and the respective tour length for one TSP. • Not able to illustrate and explain the mechanism to generate plots and compute tour lengths.	• Able to generate optimal tour and the respective tour length for two TSPs. • Able to illustrate and explain the mechanism to generate plots and compute tour lengths with minimal clarity.	• Able to generate optimal tour and the respective tour length for three TSPs. • Able to illustrate and explain the mechanism to generate plots and compute tour lengths with satisfactory clarity.	• Able to generate optimal tour and the respective tour length for four TSPs. • Able to illustrate and explain the mechanism to generate plots and compute tour lengths with good clarity.	• Able to generate optimal tour and the respective tour length for five TSPs. • Able to illustrate and explain the mechanism to generate plots and compute tour lengths with excellent clarity.
Algorithm design (40%)	• Not able to apply any new idea or knowledge to a given problem. • The algorithm implementation is not correct and not comprehensive.	• Limited ability to apply new ideas or knowledge. • The algorithm implementation is minimally correct.	• Able to apply new ideas or knowledge to a given problem. • The algorithm implementation is partially correct.	• Able to apply new ideas or knowledge to a given problem. • The algorithm implementation is correct and comprehensive.	• Able to apply new ideas or knowledge to a given problem and able to propose alternative applications. • The implementation based on the alternative applications is correct and comprehensive.

	Very Weak (1 – 2 points)	Weak (3 – 4 points)	Fair (5 – 6 points)	Good (7 – 8 points)	Very Good (9 – 10 points)
Experimental results (20%)	• Interpret and explain the results. • The explanation and the results are presented unclearly, loosely, and disorganized.	• Interpret and explain the results. • The explanation and the results are presented with minimal clarity, comprehensiveness, and organization.	• Interpret and explain the results. • The explanation and the results are presented with satisfactory clarity, comprehensiveness, and organization.	• Interpret and explain the results. • The explanation and the results are presented with good clarity, comprehensiveness, and organization.	• Interpret and explain the results. • The explanation and the results are presented with excellent clarity, comprehensiveness, and organization.
Problems and pitfalls (10%)	Not able to perform reflection.	Able to deliver a reflection report with minimal clarity.	Able to deliver a reflection report with satisfactory clarity.	Able to deliver a reflection report with good clarity.	Able to deliver a reflection report with excellent clarity.

IMPORTANT: Students who copied or plagiarized other's work or let their work be copied or plagiarized will be given an F grade. The student may be barred from sitting for final exam and reported to the university's disciplinary board.

~~END OF ASSIGNMENT 01~~